

# Shammur Absar Chowdhury

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## Academic Expertise and Research Interest

- **Coversational AI:** Speech Discourse and Turn-Taking Modeling; Atypical Interaction; Non-Verbal and Multimodal Communication; Spoken Language Assessment.
- **NLP and Foundation Models:** LLMs; Audio/Multimodal LLMs; Representation Learning; Cross-Modal Grounding; Culturally Contextualized Reasoning.
- **Inclusive Speech Technology:** Dialectal and Accented ASR; Children ASR; Multilingual and Code-Switching ASR; Speaker recognition; Model interpretability and bias analysis.

## Scholar Profile

- Google Scholar🔗: Total Citation: **2996**, h-index: **30**, i10-index: **65**

## Education

**PhD** in Information and Communication Technology

Nov 2012 – Apr 2017

- University of Trento, Italy  
Thesis Title: [Computational Modeling of Turn-Taking Dynamics in Spoken Conversations](#)🔗.  
Supervisor: Prof. Giuseppe Riccardi, University of Trento, Italy.

**Bachelor of Science in Electronics and Communication Engineering**

Jan 2007 – Dec 2010

with 2<sup>nd</sup> major in **Bachelor of Science in Computer Science**

(Convocation Feb 2011)

- BRAC University, Bangladesh  
CGPA: 3.97/4.0 (*Gold medalist and Valedictorian for getting highest CGPA among all University Graduates, 2011*).  
Thesis Title: [Implementation of Speech Recognition System for Bangla](#)🔗.  
Supervisor: Prof. Mumit Khan, BRAC University, Bangladesh

## Research/Development Experience

**Research Scientist**, Qatar Computing Research Institute, HBKU, Qatar

Oct 2021 – Present

- **Funded projects:**
  - Speech Lead for [Fonar: Arabic LLMs](#)🔗
  - Lead PI (*HBKU Flagship Research Grant*) for [NAVIA: Neurodiversity Assessment and Voice-enabled Intervention AI](#)🔗
  - PI (*QRDI*) for MediaVoice: Emotion-Aware Arabic Voice Generation for Media
- **QCRI Projects:**
  - Lead PI for [QVoice](#)🔗: Spoken Language Assessment
  - Lead PI for AURA: Audio Understanding & Representation Alignment
  - Contributor for [LLMeBench](#)🔗: Open-source (Multimodal) LLM Benchmarking Tool (5.68K pip downloads since Aug, 2024).
  - [MultimodalXplain](#)🔗: Interpreting Multimodal Models.
- **Invention Disclosure:**
  - 2026-004 A Framework for Persona-Conditioned Multi-Turn Conversations (Persona→Scenario→Dialogue→Speech). (Filed)
  - 2025-043 EverydayMMQA: A Multilingual and Multimodal Framework for Culturally Grounded Spoken Visual QA. (Filed)
  - 2025-035 Fonar: An Arabic-Centric Multimodal Generative AI Platform. (Filed)

- 2024-035 NativQA: Multilingual Culturally-Aligned Natural Query for LLMs. (Software copyright)
- 2022-057 Multilingual Word Error Rate Estimation: e-WER3. (Elected)

**Visiting Researcher**, Frederick Jelinek Memorial Summer Workshop (JHU) Jul 2022 – Aug 2022  
Project: [Multilingual and Code-Switching Speech Recognition](#)

**Postdoctoral Researcher**, Qatar Computing Research Institute, HBKU, Qatar May 2019 – Oct 2021

- Design End-to-End Dialect Identification, Dialectal and Code-switching ASR among others.
- Design Multi-label short text classification for topic extraction (English and Arabic)
- Design Prediction model for Offensive language detection (English and Arabic)

**Research Fellow**, Center for Mind/Brain Sciences, University of Trento, Trento, Italy May 2019 – Apr 2021

- Interpretability of Language Model representation for rare linguistic phenomena
- Simulation and Interpretation of ‘online’ vs ‘off-line’ processing of sentences using transformer models.

**Postdoctoral Researcher**, Center for Mind/Brain Sciences, University of Trento, Italy Sep 2017 – Apr 2019

- Simulation of Grammaticality Judgements using DNN models
- Artificial production (Data generation) and Corrections of ungrammatical scenarios.

**PhD Researcher**, Signals and Interactive Systems Lab, University of Trento, Italy Nov 2012 – Apr 2017

- Modeling Turn-taking behavior and discourse of overlap and silence in spoken conversations (using acoustic and lexical information).
- Modeling Automatic Conversation Flow facilitator and Information profiler for call-center data analytics system.
- Extraction of Non-verbal features for para-semantic prediction (part of EU-funded Project: [SENSEI](#)).
- Cross-Language Annotation porting using crowdsourcing platform (part of EU-funded Project: [PortDial](#)).
- Speech Recognition system for Dyadic conversation – Italian (Telephony data).

**Research Assistant**, [SIS Lab](#), University of Trento, Trento, Italy Mar 2012 – Oct 2012

- Speech Recognition for Italian language using LUNA spoken dialog corpus.

**Senior Research Assistant**, Center for Research on Bangla Language Processing (CRBLP), BRAC University, Dhaka Bangladesh Oct 2010 – Feb 2012

- Developed a new voice for Bangla Text to Speech (TTS);
- Developed a newspaper speech corpus ( <https://sourceforge.net/projects/blp>).

## Teaching Experience

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**Lecturer**, Department of Computer Science and Engineering Jan 2011 – Feb 2012  
BRAC University, Dhaka Bangladesh

- Courses Taken: Introduction to Computer Science, Programming Language I and II, Numerical Methods, Electronic Device II.

**Teaching Assistant**, Department of Computer Science and Engineering Oct 2008 – Aug 2010  
BRAC University, Dhaka Bangladesh

- Assisted Courses: Programming Language I. Programming Language II.

## Leadership/Mentoring Experience

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Currently mentoring **5** students (1 RA, 3 Student Contributors, 1 Intern) across projects like Fanar, QVoice, MultimodalXplain, and AURA. Previously supervised **16** students through various QCRI RA/internship programs and other collaborations.

## Selected Projects

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### Current Projects

- **Fanar: Arabic Large Language Model Family** [↗](#) – *Speech Understanding Lead*
  - Leading the design of audio LLMs for dialectal Arabic, focusing on spoken language understanding and multimodal feedback generation. (*Part of Fanar, sponsored by Ministry of Communications and Information Technology, Qatar*)
- **NAVIA: Neurodiversity Assessment and Voice-enabled Intervention AI** [↗](#) – *Lead PI*
  - Design a holistic framework for early screening and intervention of Autism Spectrum Disorder (ASD) using multimodal AI. (*Funded by Flagship Research Grant, HBKU. HBKU-OVPR-FRG-03-3*)
- **QVoice** [↗](#) – *Lead PI*
  - Developing spoken Arabic language learning platforms with voice-enabled input and interactive feedback systems to support native and non-native learners.
- **LLMeBench** [↗](#)
  - Designing a comprehensive benchmarking framework to evaluate the understanding and generation capabilities of (multimodal) LLMs.
- **AURA: Audio Understanding & Representation Alignment** [↗](#) – *Lead*
  - Explores self-supervised and multimodal learning techniques to develop robust audio representations for audio understanding tasks. Focuses on aligning acoustic, semantic and paralinguistic features to improve cross-modal learning, and performance across multilingual, multi-dialectal and low-resource settings.
- **MultimodalXplain** [↗](#)
  - Explores how models encode and represent information across multimodal inputs (e.g., speech + text), designing methods to analyze internal representations, captured concepts, and cross-modal feature alignment.

### Past Projects

- Robust Speech Recognition for Code-switched Arabic speech
- **ASAD: Arabic Social media Analytics and unDerstanding** [↗](#)
- **Automatic Persona Generation** [↗](#)
- **TREiL: Technologies for Research and Education in Linguistics** [↗](#) (PRIN, MUR funded project, Trento, Italy).
- **SENSEI: Making Sense of Human - Human Conversation** [↗](#) (EU-funded project).
- **PortDial: Language Resources for Portable Multilingual Spoken Dialogue Systems** [↗](#) (EU-funded project).
- Conversational and Italian Telephony Speech Recognition System.
- Turn-taking modeling, Overlapping Speech and its Discourse in multi-speaker conversation
- Voice Creation for Bangla Text To Speech

### Platforms

- **MyVoice** [↗](#): Platform for collecting dialectal and accented speech data. (Co-Founder)
- **BNLP** [↗](#): Initiative for Bangla Language Computing resources. (Co-Founder)
- **ArabicSpeech** [↗](#): Platform dedicated to advancing Arabic Speech Science and Technology. (Primary Contributor)

## Selected Grants

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- *Emotion-Aware Arabic Voice Generation for Media (MediaVoice)*, Shammur Chowdhury (PI), funded by *QRDI* for 2 years, USD 81,940 (298,744.88 QAR), (2026-2028)
- *Neurodiversity Assessment and Voice-enabled Intervention AI (NAVIA)* [↗](#), Shammur Chowdhury (LPI), funded by *HBKU-OVPR*, Project Id: *HBKU-OVPR-FRG-03-09*, for 2 years, USD 207,000 (750,000 QAR), (2025-2027)
- *Native, Local and Cultural Alignment in Multilingual and Multimodal LLMs* [↗](#), Shammur Chowdhury (Co-PI), funded jointly by *AWS BOT* (USD 50,000) and *QCRI* (2025-2026)

- *Exploration of Audio Discrete tokens for multilingual models*, Shammur Chowdhury (LPI), funded in-part by Google for Education (USD 5000).
- *RNN-based Artificial Grammaticality to Probe Language Universals*, Shammur Chowdhury (Co-PI), supported by NVIDIA Academic Hardware Grant, 2018.

## Selected Talks

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- **Tutorial:** Multilingual and Multimodal LLMs in the Wild: Building for Low-Resource Languages (LREC 2026)
- **Tutorial:** LLMs for Low Resource Languages in Multilingual, Multimodal and Dialectal Settings (EACL 2024)
- **Webinar Talk:** Towards L1-aware Multilingual Mispronunciation Detection Modeling, (Speech and Language Technology in Education webinar series, 2024)
- **Invited Talk:** Learning Pronunciation with Voice-Enabled AI, (WISE Summit 2023)
- **Invited Talk:** Multilingual and Code-Switching Speech Recognition, JSALT report. (SLT 2022)
- **Invited Talk:** Fine-Grained Interpretation of End-to-End Speech Models (SIGML 2021)
- Fireside Chat on Bangla NLP (Bengali.AI 2021)

## Membership in Professional Societies

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- Member, **IEEE**, since 2009;
- Member, **ISCA** (International Speech Communication Association), since 2013;
- Member, **ACL** (Association for Computational Linguistics), since 2018.

## Professional Activities

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### Scientific Program Committee (selected 2024-2026)

**Reviewer:** ICASSP, INTERSPEECH, Computer Speech and Language, Speech Communication, The Journal of the Acoustical Society of America, ACL-AAR, ACL, COLING, NAACL, EMNLP, Language Resources and Evaluation, ACM Transactions on Asian and Low-Resource Language Information Processing, ICLR and NeurIPS.

**Associate Editor:** ACM TALIP

**Session Chair:** INTERSPEECH - 2026, 2025; EMNLP - 2023

**Area Chair:** INTERSPEECH - 2024; LREC - 2026

### Co-organizing: Conference/Workshop/Shared Tasks (selected 2022 -)

- **Shared Task Organizer:** SemEval MMCultureQA 2027 – Multilingual, multimodal, culturally grounded visual question answering, at ACL 2027, Kyoto, Japan [🔗](#).
- **Shared Task Organizer:** IQRA – Automatic Pronunciation Assessment for Modern Standard Arabic (MSA), Interspeech 2026, Sydney, Australia [🔗](#)
- **Shared Task Organizer:** ArabicNLP Task 6: ImageEval 2026 – Culturally Grounded Arabic Multimodal Evaluation, at EMNLP 2026, Budapest, Hungary [🔗](#).
- **Technical Chair:** 16th International Workshop on Spoken Dialogue Systems Technology (IWSDS 2026) [🔗](#).
- **Organizer:** Second Bangla Language Processing (BLP) @ AACL 2025 [🔗](#).
- **Organizer:** Workshop on Detecting AI Generated Content (GenAI) @ COLING 2025 [🔗](#). 30+ participants attended in-person.
- **Organizer:** First Bangla Language Processing (BLP) @ EMNLP 2023 [🔗](#). 30+ participants attended in-person and online.
- **Organizer/Event Coordinator:** Connecting Early Career Researchers - Lecture and Panels (with ISCA-PECRAC) at Interspeech 2022, 2023, 2024, and 2025.
- **Local Chair:** SLT 2022, hosted in Qatar and in the region for the first time, 200 participants registered.
- **Local Organizer:** Speech and Language Technology hackathon (SLT-CODE) 2022, Doha Qatar, with 100 students and 30 mentors.

- **Local Organizer:** Data, Technologies, and Benchmarks for the Spoken Languages of the World meeting, at SLT 2022, Qatar.
- **Shared Task Organizer:** SemEval2022 Task 3: Presupposed Taxonomies - Evaluating Neural-network Semantics (PreTENS), at NAACL 2022, USA.

## Special Interest Groups

- Speech and Language in Technology in Education, (ISCA-SLaTE)
- Postdoc & Early Career Researcher Advising Committee (ISCA-PECRAC), Coordinator
- WomenInSpeech (ISCA) and SIGML (ISCA)

## Publications

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### 2026

1. Ali, Z. S., Mubarak, H., Jung, S.-G., Bhatti, H. H., Alam, F., and Chowdhury, S. A. (2026b). Wasil: In-the-wild arabic spoken interactions with llms. In *Proc. of Interspeech*
2. Ahmed, S. F. and Chowdhury, S. A. (2026). Light-weight pronunciation assessment via discrete speech token surprisal. In *Proc. of Interspeech*
3. Mousi, B., Dalvi, F., Chowdhury, S., Alam, F., and Durrani, N. (2026a). Said aloud, read different: Cross-modal instability in multimodal models. In *Proc. of Interspeech*
4. Kheir, Y. E., Meghanani, A., Shahin, M., Ibrahim, O., Chowdhury, S. A., AlMarwani, N., Elshahawy, Y., and Ali, A. (2026). Iqra 2026: Interspeech challenge on automatic pronunciation assessment for modern standard arabic (msa). In *Proc. of Interspeech*
5. Mousi, B., Dalvi, F., Chowdhury, S. A., Alam, F., and Durrani, N. (2026b). Once correct, still wrong: Counterfactual hallucination in multilingual vision-language models. In *In Proc. of ACL*
6. Ali, Z. S., Bhatti, H. H., Nandi, R. N., Chowdhury, S. A., and Alam, F. (2026a). Menaspeechbank: A reference voice bank with persona-conditioned multi-turn conversations for audiollms. *arXiv preprint arXiv:2602.07036*
7. El Kheir, Y., Chowdhury, S. A., and Ali, A. (2026). The complementary role of para-linguistic cues for robust pronunciation assessment. In *Proc. of 16th International Workshop on Spoken Dialogue System Technology*
8. Abbas, U., Ahmad, M. S., Ahmad, M., Al-Homaid, A., Al-Nuaimi, A., Altinisik, E., Asgari, E., Chawla, S., Chowdhury, S., Dalvi, F., et al. (2026). Fanar 2.0: Arabic generative ai stack. *arXiv preprint arXiv:2603.16397*
9. Alam, F., Bhatia, G., Laskar, S. R., and Chowdhury, S. A. (2026). Beyond llm-as-a-judge: Deterministic metrics for multilingual generative text evaluation. *arXiv preprint arXiv:2604.05083*
10. Sukhadia, V. N. and Chowdhury, S. A. (2026). Harness: Lightweight distilled arabic speech foundation models. *Proc. of Speech Language Models in Low-Resource Settings: Performance, Evaluation, and Bias Analysis*
11. Rabh, M., Boshmaf, Y., Alsabah, M., Chowdhury, S., Hefeeda, M., and Khalil, I. (2026). Callshield: Secure caller authentication over real-time audio channels. *arXiv preprint arXiv:2601.09327*
12. Bhatti, H. H., Alam, F., and Chowdhury, S. A. (2026). Multi-task instruction tuning via data scheduling for low-resource arabic audiollms. *arXiv preprint arXiv:2601.12494*
13. Lachemat, H. E.-O. and Chowdhury, S. A. Child ASR adaptation with adult retention: An empirical study. In *Submitted to ArabicNLP 2026*

### 2025

14. Alam, F., Shahroor, A. E., Hasan, M. A., Ali, Z. S., Bhatti, H. H., Kmainasi, M. B., Chowdhury, S. A., Mousi, B., Dalvi, F., Durrani, N., and Milic-Frayling, N. (2025b). EverydayMMQA: A multilingual and multimodal framework for culturally grounded Spoken Visual QA. *arXiv preprint arXiv:2510.06371*

15. Chowdhury, S., Ali, A., Abdelali, A., Elfeky, M., and Darwish, K. (2025a). Unlocking the potential of arabic voice-generation technologies. *Commun. ACM*, 68(10):45–47
16. Ersoy, A., Mousi, B., Chowdhury, S., Alam, F., Dalvi, F., and Durrani, N. (2025). From words to waves: Analyzing concept formation in speech and text-based foundation models. In *Proc. of INTERSPEECH*
17. Alam, F., Arid Hasan, M., and Absar Chowdhury, S. (2025a). SpokenNativQA: Multilingual everyday spoken queries for llms. In *Proc. of INTERSPEECH*
18. Lodagala, V. S., Alkanhal, L., Izham, D., Mehta, S., Alharbi, S., Chowdhury, S., Makki, A., Hussein, H. S., Henter, G. E., and Ali, A. (2025). SawtArabi: A benchmark corpus for arabic tts. standard, dialectal and code-switching. In *Proc. of INTERSPEECH*
19. Hasan, M. A., Hasanain, M., Ahmad, F., Laskar, S. R., Upadhyay, S., Sukhadia, V. N., Kutlu, M., Chowdhury, S. A., and Alam, F. (2025). NativQA: Multilingual culturally-aligned natural query for llms. In *Proc. of ACL*
20. Lachemat, H. E.-O., Abbas, A., Oukas, N., Elkheir, Y., Haboussi, S., and Chowdhury, S. A. (2025). Cafe: A novel code-switching dataset for algerian dialect, french, and english. *Data in Brief*
21. Team, F., Abbas, U., Ahmad, M. S., Alam, F., Altinisik, E., Asgari, E., Boshmaf, Y., Boughorbel, S., Chawla, S., Chowdhury, S., et al. (2025). Fanar: An arabic-centric multimodal generative ai platform. *arXiv preprint arXiv:2501.13944*
22. Basher, M. J. I., Kowsher, M., Islam, M. S., Nandi, R. N., Prottasha, N. J., Menon, M. H., Muntasir, T. A., Chowdhury, S. A., Alam, F., Yousefi, N., et al. (2025). BnTTS: Few-shot speaker adaptation in low-resource setting. In *Proc. of NAACL*
23. Chowdhury, S. A., Almerexhi, H., Kutlu, M., Keles, K. E., Ahmad, F., Mohiuddin, T., Mikros, G., and Alam, F. (2025b). GenAI content detection task 2: Ai vs. human–academic essay authenticity challenge. In *Proc. of COLING*
24. Mousi, B., Durrani, N., Ahmad, F., Hasan, M. A., Hasanain, M., Kabbani, T., Dalvi, F., Chowdhury, S. A., and Alam, F. (2025). AraDiCE: Benchmarks for dialectal and cultural capabilities in llms. In *Proc. of COLING*

## 2024

25. Sukhadia, V. N. and Chowdhury, S. A. (2024). Children’s speech recognition through discrete token enhancement. In *Proc. of INTERSPEECH*
26. Kheir, Y. E., Mubarak, H., Ali, A., and Chowdhury, S. A. (2024). Beyond orthography: Automatic recovery of short vowels and dialectal sounds in arabic. In *Proc. of ACL*
27. El Kheir, Y., Chowdhury, S. A., and Ali, A. (2024b). L1-aware multilingual mispronunciation detection framework. In *Proc. of ICASSP*. IEEE
28. Hussein, A., Zeinali, D., Klejch, O., Wiesner, M., Yan, B., Chowdhury, S., Ali, A., Watanabe, S., and Khudanpur, S. (2024). Speech collage: code-switched audio generation by collaging monolingual corpora. In *Proc. of ICASSP*. IEEE
29. El Kheir, Y., Ali, A., and Chowdhury, S. A. (2024a). Speech representation analysis based on inter-and intra-model similarities. In *Proc. of the Explainable Machine Learning for Speech and Audio Workshop, ICASSP*. IEEE
30. Abdelali, A., Mubarak, H., Chowdhury, S. A., Hasanain, M., Mousi, B., Boughorbel, S., Abdaljalil, S., Kheir, Y. E., Izham, D., Dalvi, F., Hawasly, M., Nazar, N., Elshahawy, Y., Ali, A., Durrani, N., Milic-Frayling, N., and Alam, F. (2024). LAraBench: Benchmarking Arabic AI with Large Language Models. In *Proc. of EACL*
31. Dalvi, F., Hasanain, M., Boughorbel, S., Mousi, B., Abdaljalil, S., Nazar, N., Abdelali, A., Chowdhury, S. A., Mubarak, H., Ali, A., Hawasly, M., Durrani, N., and Alam, F. (2024). LLMeBench: A flexible framework for accelerating llms benchmarking. In *Proc. of EACL*

## 2023

32. Kheir, Y., Ali, A., and Chowdhury, S. (2023a). Automatic pronunciation assessment-a review. In *Proc. of EMNLP*
33. Kheir, Y. E., Chowdhury, S. A., and Ali, A. (2023c). Speechblender: Speech augmentation framework for mispronunciation data generation. In *Proc. of the Speech and Language Technology in Education (SLaTE)*
34. Kheir, Y. E., Chowdhury, S. A., and Ali, A. (2023b). Multi-view multi-task representation learning for mispronunciation detection. In *Proc. of the Speech and Language Technology in Education (SLaTE)*
35. Chowdhury, S. A., Durrani, N., and Ali, A. (2023). What do end-to-end speech models learn about speaker, language and channel information? a layer-wise and neuron-level analysis. *Computer Speech & Language*, 83:101539
36. Nandi, R. N., Menon, M., Muntasir, T., Sarker, S., Muhtaseem, Q. S., Islam, M. T., Chowdhury, S., and Alam, F. (2023). Pseudo-labeling for domain-agnostic bangla automatic speech recognition. In *Proc. of the First Workshop on Bangla Language Processing, BLP*
37. Chowdhury, S. A. and Ali, A. (2023). Multilingual word error rate estimation: E-wer3. In *Proc. of ICASSP*
38. Kheir, Y. E., Khnaisser, F., Chowdhury, S. A., Mubarak, H., Afzal, S., and Ali, A. (2023d). Qvoice: Arabic speech pronunciation learning application. In *Proc. of INTERSPEECH*
39. Elshahawy, Y., Kheir, Y. E., Chowdhury, S. A., and Ali, A. (2023). Myvoice: Arabic speech resource collaboration platform. In *Proc. of INTERSPEECH*
40. Mubarak, H., Hassan, S., and Chowdhury, S. A. (2023). Emojis as anchors to detect arabic offensive language and hate speech. *Natural Language Engineering (NLE) Journal*

## 2022

41. Bayerl, S. P., Roccabruna, G., Chowdhury, S. A., Ciulli, T., Danieli, M., Riedhammer, K., and Riccardi, G. (2022). What can speech and language tell us about the working alliance in psychotherapy. In *Proc. of INTERSPEECH*
42. Zamparelli, R., Chowdhury, S., Brunato, D., Chesi, C., Dell’Orletta, F., Hasan, M. A., and Venturi, G. (2022). Semeval-2022 task 3: Pretens-evaluating neural networks on presuppositional semantic knowledge. In *Proc. of SemEval*
43. Hamed, I., Hussein, A., Chellah, O., Chowdhury, S., Mubarak, H., Sitaram, S., Habash, N., and Ali, A. (2022). Benchmarking evaluation metrics for code-switching automatic speech recognition. In *Proc. of SLT. IEEE*
44. Hussein, A., Chowdhury, S. A., Abdelali, A., Dehak, N., Ali, A., and Khudanpur, S. (2022). Textual data augmentation for arabic-english code-switching speech recognition. In *Proc. of SLT. IEEE*

## 2021

45. Chowdhury, S. A., Hussein, A., Abdelali, A., and Ali, A. (2021). Towards One Model to Rule All: Multilingual Strategy for Dialectal Code-Switching Arabic ASR. In *Proc. of INTERSPEECH*
46. Ali, A., Chowdhury, S., Hussein, A., and Yasser, H. (2021b). Arabic Code-Switching Speech Recognition using Monolingual Data. In *Proc. of INTERSPEECH*
47. Mubarak, H., Hussein, A., Chowdhury, S. A., and Ali, A. (2021). QASR: QCRI Aljazeera Speech Resource. A Large Scale Annotated Arabic Speech Corpus. In *Proc. of ACL*
48. Ali, A., Chowdhury, S., Afify, M., El-Hajj, W., Hajj, H., Abbas, M., Hussein, A., Ghneim, N., Abushariah, M., and Alqudah, A. (2021a). Connecting Arabs: bridging the gap in dialectal speech recognition. *Commun, ACM*, 64(4):124–129. Publisher: ACM New York, NY, USA
49. Alam, F., Hasan, A., Alam, T., Khan, A., Tajrin, J., Khan, N., and Absar Chowdhury, S. (2021). A review of bangla natural language processing tasks and the utility of transformer models. *arXiv e-prints*, pages arXiv–2107
50. Hussein, A., Chowdhury, S., and Ali, A. (2021). Balanced end-to-end monolingual pre-training for low-resourced indic languages code-switching speech recognition. *arXiv preprint arXiv:2106.05885*

51. Salminen, J., Jung, S.-G., Chowdhury, S., Robillos, D. R., and Jansen, B. (2021). The ability of personas: An empirical evaluation of altering incorrect preconceptions about users. *International Journal of Human-Computer Studies*, 153:102645. Publisher: Academic Press
52. Jansen, B. J., Jung, S.-g., Chowdhury, S. A., and Salminen, J. (2021). Persona analytics: Analyzing the stability of online segments and content interests over time using non-negative matrix factorization. *Expert Systems with Applications*, 185:115611

## 2020

53. Chowdhury, S. A., Ali, A., Shon, S., and Glass, J. (2020b). What does an end-to-end dialect identification model learn about non-dialectal information? In *Proc. of INTERSPEECH*
54. Chowdhury, S. A., Samih, Y., Eldesouki, M., and Ali, A. (2020d). Effects of dialectal code-switching on speech modules: A study using egyptian arabic broadcast speech. In *Proc. of INTERSPEECH*
55. Chowdhury, S. A., Mubarak, H., Abdelali, A., Jung, S.-g., Jansen, B. J., and Salminen, J. (2020c). A multi-platform arabic news comment dataset for offensive language detection. In *Proc. of LREC*
56. Chowdhury, S. A., Abdelali, A., Darwish, K., Jung, S.-g., Salminen, J., and Jansen, B. J. (2020a). Improving arabic text categorization using transformer training diversification. In *Proc. of WANLP, COLING*
57. Salminen, J., Jung, S.-G., Chowdhury, S., Sengün, S., and Jansen, B. J. (2020e). Personas and analytics: A comparative user study of efficiency and effectiveness for a user identification task. In *Proc. of CHI*, pages 1–13
58. Salminen, J., Hopf, M., Chowdhury, S. A., Jung, S.-g., Almerexhi, H., and Jansen, B. J. (2020c). Developing an online hate classifier for multiple social media platforms. *Human-centric Computing and Information Sciences*, 10(1):1
59. Salminen, J., Guan, K., Jung, S.-G., Chowdhury, S. A., and Jansen, B. J. (2020b). A literature review of quantitative persona creation. In *Proc. of CHI*, pages 1–14
60. Hasan, M. A., Tajrin, J., Chowdhury, S. A., and Alam, F. (2020). Sentiment classification in bangla textual content: A comparative study. In *Proc. of ICCIT*, pages 1–6. IEEE
61. Salminen, J., Rao, R. G., Jung, S.-g., Chowdhury, S. A., and Jansen, B. J. (2020h). Enriching social media personas with personality traits: A deep learning approach using the big five classes. In *Proc. of the International Conference on Human-Computer Interaction*, pages 101–120. Springer, Cham
62. Salminen, J., Jung, S.-g., Chowdhury, S. A., and Jansen, B. J. (2020f). Rethinking personas for fairness: Algorithmic transparency and accountability in data-driven personas. In *Proc. of the International Conference on Human-Computer Interaction*, pages 82–100. Springer, Cham
63. Hassan, S., Samih, Y., Mubarak, H., Abdelali, A., Rashed, A., and Chowdhury, S. A. (2020). Alt submission for osact shared task on offensive language detection. In *Proc. of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 61–65
64. Salminen, J., Jung, S.-g., Santos, J. M., Chowdhury, S., and Jansen, B. J. (2020g). The effect of experience on persona perceptions. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–9
65. Jung, S.-G., Salminen, J., Chowdhury, S. A., Ramirez Robillos, D., and Jansen, B. J. (2020). Things change: Comparing results using historical data and user testing for evaluating a recommendation task. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–7
66. Salminen, J., Jung, S.-g., Chowdhury, S., and Jansen, B. J. (2020d). Analyzing demographic bias in artificially generated facial pictures. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–8
67. Salminen, J., Vahlo, J., Koponen, A., Jung, S.-G., Chowdhury, S. A., and Jansen, B. J. (2020i). Designing prototype player personas from a game preference survey. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–8
68. Salminen, J., Froneman, W., Jung, S.-g., Chowdhury, S., and Jansen, B. J. (2020a). The ethics of data-driven personas. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–9

## 2019

69. Chowdhury, S. A. and Zamparelli, R. (2019). An lstm adaptation study of (un)grammaticality. In *Proc. of the Black-boxNLP: Analyzing and Interpreting Neural Networks for NLP, ACL*
70. Chowdhury, S. A., Stepanov, E. A., Danieli, M., and Riccardi, G. (2019). Automatic classification of speech overlaps: Feature representation and algorithms. *Computer Speech & Language*, 55:145–167
71. Hasan, A., Alam, F., Chowdhury, S. A., and Khan, N. (2019). Neural vs statistical machine translation: Revisiting the bangla-english language pair. In *Proc. of ICBSLP 2019*

## 2018

72. Chowdhury, S. A. and Zamparelli, R. (2018). Rnn simulations of grammaticality judgments on long-distance dependencies. In *Proc. of COLING*
73. Stepanov, E., Lathuiliere, S., Chowdhury, S. A., Ghosh, A., Vieri, R.-L., Sebe, N., and Riccardi, G. (2018a). Depression severity estimation from multiple modalities. In *Proc. of Healthcom, IEEE*
74. Stepanov, E. A., Chowdhury, S. A., Bayer, A. O., Ghosh, A., Klasinas, I., Calvo, M., Sanchis, E., and Riccardi, G. (2018b). Cross-language transfer of semantic annotation via targeted crowdsourcing: Task design and evaluation. *Language Resources and Evaluation Journal*
75. Chowdhury, S. A., Alam, F., and Khan, N. (2018). Towards bangla named entity recognition. In *Proc. of ICCIT. IEEE*

## 2017

76. Chowdhury, S. A., Stepanov, E., Danieli, M., and Riccardi, G. (2017b). Functions of silences towards information flow in spoken conversation. In *Proc. of the Workshop on Speech-Centric Natural Language Processing, EMNLP, Copenhagen, Denmark*
77. Chowdhury, S. A., Alam, F., Khan, N., and Noori, S. R. (2017a). Bangla grapheme to phoneme conversion using conditional random fields. In *Proc. of ICCIT. IEEE*
78. Chowdhury, S. A. and Riccardi, G. (2017). A deep learning approach to modeling competitiveness in spoken conversation. In *Proc. of ICASSP. IEEE*

## 2016

79. Alam, F., Chowdhury, S. A., Danieli, M., and Riccardi, G. (2016a). How interlocutors coordinate with each other within emotional segments? In *Proc. of COLING*
80. Alam, F., Chowdhury, S. A., and Noori, S. R. H. (2016b). Bidirectional lstms—crfs networks for bangla pos tagging. In *Proc. of ICCIT. IEEE*
81. Chowdhury, S. A., Stepanov, E., and Riccardi, G. (2016a). Predicting user satisfaction from turn-taking in spoken conversations. In *Proc. of INTERSPEECH*
82. Chowdhury, S. A., Stepanov, E. A., and Riccardi, G. (2016b). Transfer of corpus-specific dialogue act annotation to iso standard: Is it worth it? In *Proc. of LREC*
83. Riccardi, G., Stepanov, E. A., and Chowdhury, S. A. (2016). Discourse connective detection in spoken conversations. In *Proc. of ICASSP. IEEE*

## 2015

84. Stepanov, E., Favre, B., Alam, F., Chowdhury, S., Singla, K., Trione, J., Béchet, F., and Riccardi, G. (2015). Automatic summarization of call-center conversations. In *Proc. of ASRU*
85. Chowdhury, S. A., Danieli, M., and Riccardi, G. (2015c). The role of speakers and context in classifying competition in overlapping speech. In *Proc. of INTERSPEECH*

86. Chowdhury, S. A., Calvo, M., Ghosh, A., Stepanov, E. A., Bayer, A. O., Riccardi, G., García, F., and Sanchis, E. (2015a). Selection and aggregation techniques for crowdsourced semantic annotation task. In *Proc. of INTERSPEECH*
87. Chowdhury, S. A., Danieli, M., and Riccardi, G. (2015b). Annotating and categorizing competition in overlap speech. In *Proc. of ICASSP*. IEEE

#### **2014**

88. Chowdhury, S. A., Riccardi, G., and Alam, F. (2014b). Unsupervised recognition and clustering of speech overlaps in spoken conversations. In *Proc. of Workshop on Speech, Language and Audio in Multimedia*
89. Chowdhury, S. A., Ghosh, A., Stepanov, E. A., Bayer, A. O., Riccardi, G., and Klasinas, I. (2014a). Cross-Language Transfer of Semantic Annotation via Targeted Crowdsourcing. In *Proc. of INTERSPEECH*, Singapore

#### **2013**

90. Riccardi, G., Ghosh, A., Chowdhury, S. A., and Bayer, A. O. (2013). Motivational Feedback in Crowdsourcing: A Case Study in Speech Transcription. In *Proc. of INTERSPEECH*, Lyon, France

#### **2011**

91. Habib, S. M. M., Alam, F., Sultana, R., Chowdhury, S. A., and Khan, M. (2011). Phonetically balanced bangla speech corpus. In *Proc. of HLTD*, Egypt